

Tide retrospective: recovery from asymptotic data

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

The previous tide¹ recovered the degree and scale of a pure even-monomial potential from *exact* eventual equality of partition functions. The germbij note’s Section 7.4 hypothesis is weaker: agreement of asymptotic *expansions*. This tide closes that gap, so the recovery thread’s hypotheses now match the note’s.

2 The thing formalised

Two new theorems and a refactor in `Laplace/OneD/Recovery.lean`, sorry-free with zero warnings. Uniqueness of power-law asymptotes²: if $\alpha_1 t^{\beta_1} \sim \alpha_2 t^{\beta_2}$ at $t \rightarrow \infty$ with $\alpha_i > 0$, then $\beta_1 = \beta_2$ and $\alpha_1 = \alpha_2$. And the asymptotic recovery theorem³: asymptotic equivalence of the partition functions of $a_i x^{2k_i}/(2k_i)!$ already forces $k_1 = k_2$ and $a_1 = a_2$. Per the consult’s advice, the closed-form power representation, the coefficient positivity, and the final algebraic extraction were factored out of the exact-equality theorem into three shared lemmas⁴, so the exact and asymptotic recovery proofs are each now a few lines over a common tail.

3 Proof strategy

The equivalence unfolds to the ratio tending to 1 (`isEquivalent_iff_tendsto_one`; the denominator is eventually nonzero since t is eventually positive). The ratio is eventually the function $(\alpha_1/\alpha_2) t^{\beta_1 - \beta_2}$, and trichotomy on $\beta_1 - \beta_2$ finishes: a positive gap sends the ratio to infinity (contradicting a finite limit), a negative gap sends it to zero (contradicting $1 = 0$ by uniqueness of limits), and a zero gap makes the ratio the constant α_1/α_2 , which uniqueness of limits pins to 1. The corollary transports the equivalence along the eventual equality with the closed forms (`isEquivalent.congr_left/congr_right`) and reuses the shared extraction.

4 Roadblocks and resolutions

Three build iterations, all elaboration-level. The ratio that `isEquivalent_iff_tendsto_one` produces is a `Pi.div` of `lambdas`, which `field_simp` will not see through until `Pi.div_apply` is unfolded — the same class as the repo’s documented `Pi.add` gotcha, now observed for division. The

¹2026-08-09-09-00-tide-germbij-recovery-1d.

²`power_asymptote_unique`.

³`kth_partitionFunction_recovery_of_isEquivalent`.

⁴`partitionFunction_kth_smul_rpow`, `kth_coeff_pos`, `kth_recovery_of_data`.

`rpow` limit lemmas are top-level constants, not in the `Real` namespace. And `tendsto_nhds_unique` against a constant function needs the constant-limit witness stated with an explicit type ascription, or the elaborator unifies the function the wrong way.

5 What was Mathlib, what was new

Mathlib: the equivalence-to-ratio characterisation, the `rpow` limits at infinity, uniqueness of limits, and the congruence lemmas for `IsEquivalent`. New: only the trichotomy packaging and the factoring; the seabed's closed form continues to carry the content.

6 Lessons

`Pi.div` joins `Pi.add` in the `unfold-before-simp` gotcha class; promoted to the repo `CLAUDE.md` alongside it. When a deliberated proof plan names limit lemmas, checking their namespaces in the same probe as their existence costs nothing and saves an iteration.

7 Follow-ups

- The genuine next depth in the recovery thread: subleading coefficients. The note recovers each a_j of $ax^{2k} + \sum_{j>2k} a_j x^j$ inductively from corrections at order $t^{-(j-2k)/(2k)}$; a first Lean step would be the single-correction case $ax^{2k} + bx^{2k+2}$, where the seabed's moment closed forms should express the first correction coefficient explicitly in b .
- Tag the note's Section 7.4 dimension-one item with the asymptotic theorem once merged (it certifies the expansion-level hypothesis).